

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 15: Similarity (With Applications to Maps and Models)

Previous year board questions, topic-wise: 4 questions

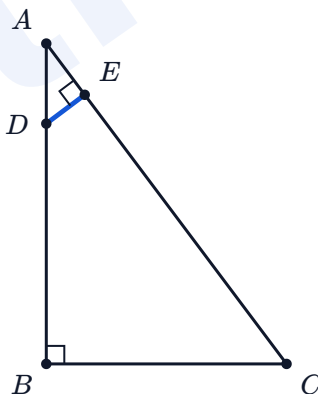
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Similar triangles

1. $\triangle ABC \sim \triangle EFG$ with $\angle ABC = \angle EFG = 60^\circ$, $AB = 15$ cm, $BC = 3$ cm and $EF = 75$ cm. Find FG . (a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm **ICSE 2025, 1 mark**

2 Maps and models

2. In $\triangle ABC$, $\angle ABC = 90^\circ$, $AB = 20$ cm and $AC = 25$ cm. D is a point on AB and E is a point on AC with $DE \perp AC$ and $DE = 3$ cm. (a) Prove that $\triangle ABC \sim \triangle AED$. (b) Find BC , AD and AE . (c) $BCED$ is a plot of land drawn on a map, and its actual area is 576 m². Find the scale factor of the map. **ICSE 2025, 4 marks**



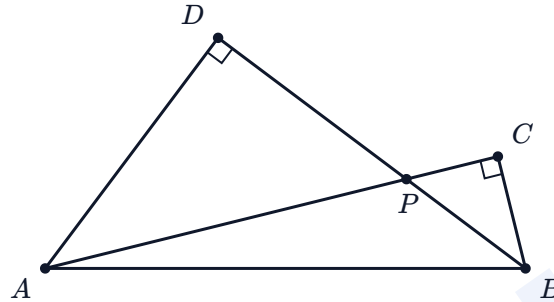
3 Conditions for similarity

3. $\triangle ABC \sim \triangle PQR$. AD bisects $\angle BAC$ and meets BC at D ; PS bisects $\angle QPR$ and meets QR at S . Which statement is true? (a) $\triangle ABC \sim \triangle PQS$ (b) $\triangle ABD \sim \triangle PQS$ (c) $\triangle ABD \sim \triangle PSR$ (d) $\triangle ABC \sim \triangle PSR$ **ICSE 2024, 1 mark**

4 Areas of similar triangles

4. $\triangle ADB$ and $\triangle ACB$ are right-angled at D and C , on the same side of AB , and AC meets BD at P . $AB = 10$ cm, $AD = 6$ cm, $BC = 2.4$ cm and $DP = 4.5$ cm. (a) Prove that $\triangle APD \sim \triangle BPC$. (b) Find BD and PB . (c) Hence find PA . (d) Find area of $\triangle APD$: area of $\triangle BPC$.

ICSE 2024, 4 marks



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